[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

DRILLING MACHINE, UPRIGHT, FLOOR-MOUNTED, ROUND AND BOX COLUMN, POWER-FEED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers power-feed, round and box column, floor-mounted, drilling machines intended for a variety of drilling operations in production shops.

2. CLASSIFICATION. The drilling machines shall be of the following types, classes, styles, and sizes. The type, class, style, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Round or box upper column, round lower column (style A only)
Type II - Round or box upper column, box lower column

Class 1 - Sliding drill head, fixed worktable
Class 2 - Sliding drill head, adjustable worktable
Class 3 - Fixed drill head, adjustable worktable

Style A - One spindle
Style B - Two spindles
Style C - Three spindles
Style D - Four spindles
Style E - Five spindles
Style F - Six spindles

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3413

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Size 1 - 1/4-inch diameter
Size 2 - 1/2-inch diameter
Size 3 - 3/4-inch diameter
Size 4 - 1-inch diameter
Size 5 - 1 1/4-inch diameter
Size 6 - 1 1/2-inch diameter
Size 7 - 1 3/4-inch diameter
Size 8 - 2-inch diameter
Size 9 - 2 1/2-inch diameter
Size 10 - 3-inch diameter
Size 11 - 3 1/2-inch diameter

3. SALIENT CHARACTERISTICS

3.1 General requirements. The drilling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The drilling machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall incorporate materials, shape, and mass to withstand stress and loads and to minimize vibrations developed during operation.

3.2.2 Worktable. Unless otherwise specified (see 7.2(c)), the worktable dimensions and weight holding capacity shall be that considered as standard by the manufacturer. The worktable shall have the ability to catch coolant and return it to the coolant sump. The number and size of T-slots shall be as specified (see 7.2(d)). The dimensions of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) Standards B5.1, "T-Slots, Their Bolts, Nuts, and Tongues". Class 1 drilling machines shall have a fixed worktable. Class 2 and class 3 drilling machines shall have a vertically adjustable worktable with height adjustment accomplished by either hand crank or powered means as specified (see 7.2(e)).

3.2.3 Column. Type I machines shall have a round or box upper column with a round lower column. Type II machines shall have a round or box upper column with a box lower column. The upper columns on machine style B through style F shall be identical and interchangeable with respect to location.

3.2.4 Drill head assembly. The drill head assembly shall consist of the power transmission and spindle and quill. The drill head assembly on class 1 and class 2 drilling machines shall be vertically adjustable along the column. The drill head assembly on class 3 drilling machines shall be mounted in a fixed position.

3.2.4.1 Power transmission. Unless otherwise specified (see 7.2(f)), the power transmission shall provide multiple speed changes and shall be of a design considered standard by the manufacturer.

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3.2.4.2 Spindle and quill. Manual and power feed shall be provided for the quill and spindle assembly. The spindle and quill shall be counterbalanced and retract automatically when the feed is released. When specified (see 7.2(g)), a lead screw and nut shall be provided for tapping operations. The quill shall have means for support and alignment throughout the full travel of the quill. When specified (see 7.2(h)), the spindle shall be capable of clockwise and counterclockwise rotation. Unless otherwise specified (see 7.2(i)), an adjustable depth stop with a positive locking device shall be provided. When the preset depth is reached the spindle and quill shall retract automatically. The depth stop shall have an indicating scale divided into increments no greater than 1/8-inch.

3.2.4.3 Spindle nose taper. Unless otherwise specified (see 7.2(j)), the spindle nose shall have a taper considered standard by the manufacturer to accept drill shanks or chucks. The taper shall conform to ANSI Standard B5.10, "Machine Tapers". When specified (see 7.2(k)), a chuck shall be furnished with the drilling machine.

3.2.5 Controls. The controls shall be located for convenient access by the operator. Unless otherwise specified (see 7.2(l)), the following controls, as a minimum, shall be provided:

a. On/off power switch
b. Emergency stop button
c. Coolant flow
d. Spindle rotation direction selector
e. Motor speed selector
f. Power feed and rate selector

3.2.6 Lubrication system. Unless otherwise specified (see 7.2(m)), the drilling machine shall be equipped with an automatic lubricating system to direct oil flow from the reservoir to all machine components requiring continuous lubrication. Means for determining oil level in the reservoir shall be provided. The lubricating system shall include a cleanable or replaceable filter. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.7 Coolant system. Unless otherwise specified (see 7.2(n)), the drilling machine shall be provided with a flood-type coolant system. As a minimum, the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The machine shall be equipped with a system to control the rate of clean coolant flow from the reservoir and to direct the coolant flow to nozzles mounted on the drill heads. The system shall have sufficient capacity to provide full cooling flow to all drill heads simultaneously. When specified (see 7.2(o)), an optional filtration system shall be provided.

3.2.8 Hydraulic system. When provided, the hydraulic system shall conform to the requirements of National Fluid Power Association (NFPA) Standard T2.24.1, "Hydraulic Fluid Power - System Standard for Stationary Industrial Machinery - First Edition". The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and controls. Means shall be provided to

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prevent damage to the components from overpressure. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.2.9 Electrical system. Unless otherwise specified (see 7.2(p)), the electrical system shall conform to the ANSI/National Fire Protection Association (NFPA) Standard 79, "Electrical Standards for Industrial Machinery". Electric motors of one horsepower or less shall operate from a dual 115/230-volt, single-phase, 60-hertz circuit; and electric motors of more than one horsepower shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit. The machine shall be initially wired for the voltage specified (see 7.2(q)). Unless otherwise specified (see 7.2(r)), the control system shall operate on a 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.10 Electric motors. The drilling machine drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to National Electrical Manufacturers Association (NEMA) Standard MG-1, "Motors and Generators". The controls and motors shall conform to NEMA ICS-1 and ICS-3, "Industrial Control and Systems: Factory Built Assemblies". Unless otherwise specified (see 7.2(s)), the motors shall have open dripproof enclosures.

3.2.11 Optional accessories. Optional accessories shall be provided as specified (see 7.2(t)).

3.3 Performance. Unless otherwise specified (see 7.2(u)), the drilling machine shall meet the alignment and performance requirements shown in tables I and II, respectively, when tested in accordance with sections 5.3 through 5.4.

TABLE I. Alignment requirements.

Parameter	Maximum allowable deviation
Working surface of base flatness, width and length	0.0016 inch per foot, 0.01 inch maximum
Spindle runout, 1 inch from nose	0.001 inch TIR
Spindle square with worktable (at maximum nose-to-table distance), longitudinal and transverse planes	0.003 inch per foot (nose-to-table distance)

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TABLE II. Performance requirements.

Parameter	Tolerance
Vertical alignment	0.010 inch/foot
Hole straightness	0.001 inch from hole top to bottom
Hole maximum diameter	Test drill diameter +0.007 inch
Out of roundness	0.003 inch

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(v)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI Standard B11.8, "Machine Tools - Drilling, Milling, and Boring Machines - Safety Requirements for Construction, Care, and Use". The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) Standard 29 CFR PART 1910, "General Industry, Safety and Health Standards". If a conflict arises between ANSI B11.8 and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(w)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(x)), measuring and indicating devices shall be graduated in both the US and SI systems of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(y)), the nameplate shall contain the following information:

Nomenclature
Manufacturer's name
Machine model designation
Power input (volts, total amperes, phase, frequency)
Short-circuit/over-current rating
Contract or order number
National stock number or plant equipment code
Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(z)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

Points of lubricant application.
Servicing interval.
Type of lubricant(s) with SAE number or lubricant identifier.

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product as that which the supplier offers for sale on the commercial market. The government reserves the right to require proof of such conformance.

5.2 Inspection. The drilling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(aa)), the drilling machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(bb)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(cc)). Unless otherwise specified (see 7.2(dd)), the machine shall be tested in accordance with environmental conditions of ambient temperature between 41 degrees and 104 degrees Fahrenheit and humidity between 20 percent and 95 percent, noncondensing.

5.3.2 Operational tests. The drilling machine shall be operated in accordance with the manufacturer's standard operating procedures. The drilling machine shall be operated at no load for not less than 1/2 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall be conducted on each spindle in accordance with sections 5.3.3.1 through 5.3.3.2. A complete test discussed in sections 5.3.3.1 and 5.3.3.2 shall be performed on each spindle. There shall be no evidence of overloading, overheating, excessive vibration, or deformation of any part of the machine during or after the test.

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5.3.3.1 Machine alignment. A machine alignment inspection shall be performed using the criteria shown in table I before and after the performance inspection.

5.3.3.2 Drill test. A test workpiece 1.0 inch thick and made of soft cast iron with a Brinell hardness of 120-150 shall be used. The test shall consist of drilling five holes. One test workpiece shall be used for each spindle. Five holes shall be drilled through the test workpiece in 12 minutes. The drill bit nominal diameter shall be the maximum rated drilling diameter for the machine. Drill bit material, drill spindle speed, and feed rate shall be at the option of the manufacturer. The holes shall be measured and shall not exceed the values shown in table II for vertical alignment, hole straightness, maximum hole diameter, and out of roundness. The diameter of all holes shall be measured from the top to the bottom in three places. Out of roundness (the difference between the largest diameter of the hole and smallest diameter of the hole) shall be determined by moving the measuring device 360 degrees around the drilled hole.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(ee)).

5.4 Acceptance. Unless otherwise specified (see 7.2(ff)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables I and II. Failure of the machine to meet these requirements shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified (7.2(gg)).

7. NOTES

7.1 Sources of documents

7.1.1 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.2 NFPA standards. Copies of NFPA (fire protection) standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101. Copies of NFPA (fluid power) standards may be obtained from the National Fluid Power Association, 3333 North Mayfair Road, Suite 311, Milwaukee, WI 53222-3219.

7.1.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.4 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Washington, DC 20210.

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7.1.5 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Type, class, style, and size of machine required (see 2)
c. Weight of workpieces, as specified (see 3.2.2)
d. Worktable T-slots, if required (see 3.2.2)
e. Worktable height adjustment for class 2 and class 3 machines, as specified (see 3.2.2)
f. Power transmission type, if different (see 3.2.4.1)
g. Lead screw and nut, if required (see 3.2.4.2)
h. Reverse spindle direction, if required (see 3.2.4.2)
i. Adjustable depth stop, if different (see 3.2.4.2)
j. Spindle nose taper, if different (see 3.2.4.3)
k. Chuck, if required (see 3.2.4.3)
l. Controls, if different (see 3.2.5)
m. Lubrication system, if different (see 3.2.6)
n. Coolant system, if different (see 3.2.7)
o. Coolant system filter, if required (see 3.2.7)
p. Electrical system per NFPA 79, if different (see 3.2.9)
q. Motor voltage, as specified (see 3.2.9)
r. Control system voltage, if different (see 3.2.9)
s. Motor enclosure, if different (see 3.2.10)
t. Optional accessories, as specified (see 3.2.11)
u. Performance, if different (see 3.3)
v. Machine dimensions and weight, as specified (see 3.4)
w. System of units required (see 3.6)
x. Dual system of units, if required (see 3.6)
y. Nameplate, if different (see 3.7)
z. Lubrication plate or chart, if required (see 3.8)
aa. Testing, if different (see 5.3)
bb. Test responsibility, if different (see 5.3.1)
cc. Certification of calibration, as specified (see 5.3.1)
dd. Test environmental conditions, if different (see 5.3.1)
ee. Optional tests, as specified (see 5.3.4)
ff. Acceptance, if different (see 5.4)
gg. Packaging requirements, as specified (see 6.1)

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7.3 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.

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- Size:
  - 1 - 1/4-inch diameter
  - 2 - 1/2-inch diameter
  - 3 - 3/4-inch diameter
  - 4 - 1-inch diameter
  - 5 - 1 1/4-inch diameter
  - 6 - 1 1/2-inch diameter
  - 7 - 1 3/4-inch diameter
  - 8 - 2-inch diameter
  - 9 - 2 1/2-inch diameter
  - 10 - 3-inch diameter
  - 11 - 3 1/2-inch diameter
- Style:
  - A - 1 spindle
  - B - 2 spindles
  - C - 3 spindles
  - D - 4 spindles
  - E - 5 spindles
  - F - 6 spindles
- Class:
  - 1 - Sliding head, fixed table
  - 2 - Sliding head, adjustable table
  - 3 - Fixed head, adjustable table
- Type:
  - A - Type I, Round or Box Upper Column, Round Lower Column
  - B - Type II, Round or Box Upper Column, Box Lower Column
- CID number

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - SM
Navy - MC, OS, YD2

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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